ChemComm

COMMUNICATION

View Article Online
View Journal | View Issue

Check for updates

Cite this: *Chem. Commun.*, 2021, 57, 9284

Received 17th June 2021,
Accepted 16th August 2021

DOI: 10.1039/d1cc03206b

rsc.li/chemcomm

Published on 16 August 2021. Downloaded by Fudan University on 1/17/2022 12:09:40 PM.

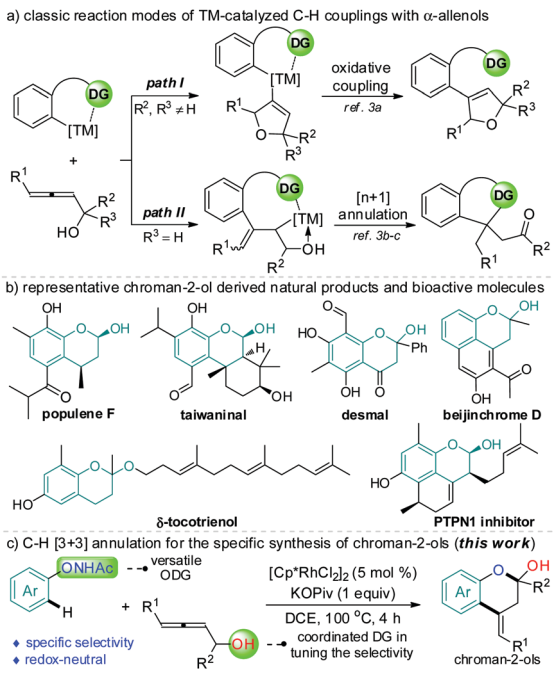
Rh(III)-Catalyzed and synergistic dual directing group-enabled redox-neutral [3+3] annulation of *N*-phenoxyacetamides with α-allenols†

Fangyuan Chen,‡ Junyuan Tang,‡ Yinhui Wei, Jingyuan Tian, Hui Gao, Wei Yi* and Zhi Zhou*

By virtue of α-allenols as innovative three-carbon annulation components, the Rh(III)-catalyzed redox-neutral C–H coupling of *N*-phenoxyacetamides with α-allenols has been realized for the assembly of 4-alkylidene chroman-2-ol frameworks *via* an unusual [3+3] annulation. This transformation features good functional group tolerance, specific regio-/chemoselectivity and potential synthetic utility. Mechanistic studies reveal that synergistic coordination modes between the dual directing groups (–ONHAc and –OH) and the rhodium metal center account for the observed exclusive selectivity.

Due to their remarkable reactivity, allenes have been proven to be versatile reagents that participate in diversified transformations including transition metal (TM)-catalysed C–H functionalization for the construction of privileged motifs from readily available raw materials.[1] Typically, the regio-/chemoselectivity control represents a major challenge for allene-involved reactions, in which both electronic and steric effects play essential roles in determining the reaction outcome.[2] As a consequence, more recently α-allenols equipped with a free hydroxyl group have been developed as reactive coupling partners (CPs) in TM-catalysed and directing group (DG)-assisted C–H annulations. The synergistic coordination modes of DGs and –OH account for the tunable reaction selectivity, giving specific access to dihydrofurans or [*n*+1] annulation products (Scheme 1a).[3] Despite these impressive advances, further exploration of novel reaction manifolds of α-allenols to furnish intriguing structures is still highly desirable.

On the other hand, *N*-phenoxy amides bearing the –ONHR moiety have been developed as powerful oxidizing directing groups (ODGs) for Rh(III)-, Ir(III)-, Co(III)-, Ru(II)- or Pd(II)-catalysed C–H functionalization,[4,5] among which several hydroxyl group-tethered CPs *e.g.* propargyl and cyclopropenyl alcohols were viable substrates to couple with *N*-phenoxy amides in a regio- and chemoselective manner.[6] Mechanistic studies implied that the synergistic coordination effect between the dual directing groups (DDGs, –ONHR and –OH) and the catalyst metal centre was the key factor for selectivity control. Encouraged by these results and in continuation of our interest in developing DDGs-enabled novel C–H functionalization,[6a,c,d,7] we would like to report herein the Rh(III)-catalysed redox-neutral C–H coupling of



Scheme 1 Diversified C–H functionalization with α-allenols.

Key Laboratory of Molecular Target & Clinical Pharmacology and the State & NMPA Key Laboratory of Respiratory Disease, School of Pharmaceutical Sciences & the Fifth Affiliated Hospital, Guangzhou Medical University, Guangzhou, Guangdong 511436, P. R. China. E-mail: yiwei@gzhmu.edu.cn, zhouzhi@gzhmu.edu.cn

† Electronic supplementary information (ESI) available: Experimental procedures, characterization of products, X-ray single-crystal diffraction data and copies of ^{1}H and ^{13}C NMR spectra. CCDC 2084815. For ESI and crystallographic data in CIF or other electronic format see DOI: 10.1039/d1cc03206b

‡ These authors contributed equally.

This journal is © The Royal Society of Chemistry 2021

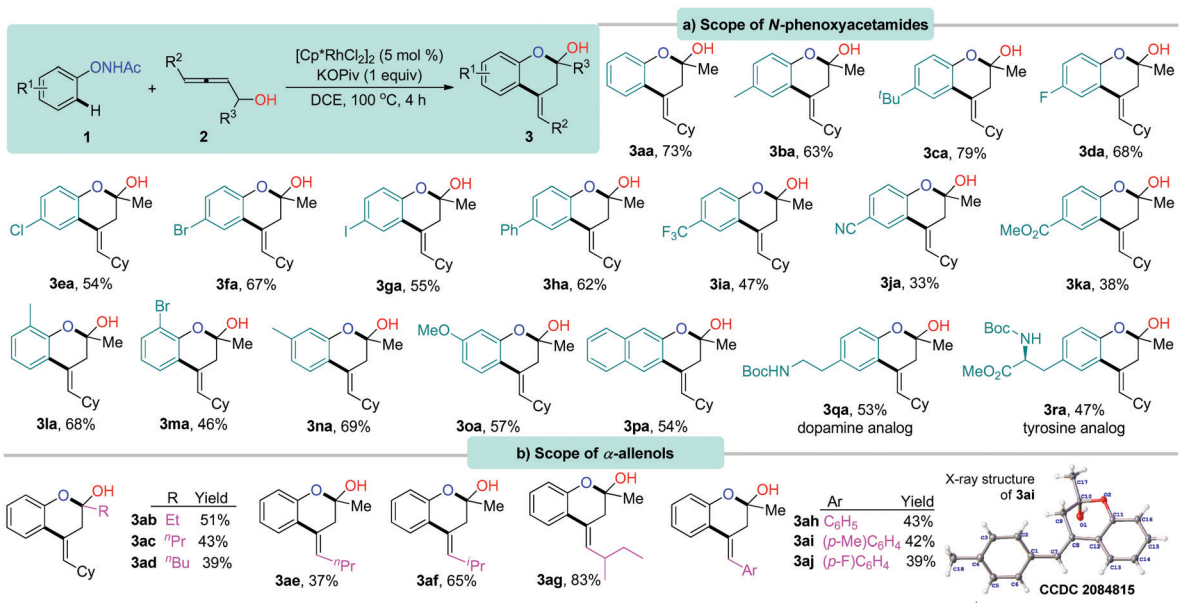
Published on 16 August 2021. Downloaded by Fudan University on 1/17/2022 12:09:40 PM.

N-phenoxyacetamides with α-allenols for the specific construction of 4-alkylidene chroman-2-ols *via* an unusual [3+3] annulation, in which the α-allenols work as innovative three-carbon annulation components (Scheme 1c). Considering the novel reaction mode, the specific selectivity control as well as the widespread existence and the potential utility of chroman-2-ol structural motifs in nature (Scheme 1b),[8] the developed methodology should be of current interest and have profound synthetic potential.

At the outset of our investigation, *N*-phenoxyacetamide **1a** was selected as the model substrate to react with α-allenol **2a** in the presence of $[Cp^*RhCl_2]_2$. The reaction proceeded smoothly in DCE with CsOAc being the additive, delivering an intriguing 4-alkylidene chroman-2-ol derivative **3aa** *via* regioselective [3+3] annulation (Table S1, entry 1, ESI†). Considering the specific regio-/chemoselectivity, the α-allenol-mediated innovative annulation mode, and the interesting 4-alkylidene chroman-2-ol structure of this transformation, we next systematically screen several experimental parameters to figure out the optimized reaction conditions (see Table S1 in the ESI† for details). Preliminary screening revealed that the reaction could be finished within 4 hours at an increased temperature, and KOPiv appeared to be an ideal additive. Most other organic solvents were also compatible but resulted in relatively lower yields compared with that of DCE. Control experiments showed that other Rh(III) species bearing different cyclopentadiene ligands and Ir(III), Ru(II) or Co(III) catalysts all gave inferior results. Various *N*-phenoxy amides equipped with different substituents on the nitrogen atom were also tested and delivered the desired product **3aa** smoothly while *N*-phenoxytosylamide proved to be ineffective.

Having established the optimal reaction conditions, we next investigated the substrate scope and compatibility for this reaction (Scheme 2). A series of *N*-phenoxyacetamides was first examined to react with α-allenol **2a**, the results revealed that various most commonly encountered functional groups including alkyl (**3ba** and **3ca**), halogen (**3da–ga**), phenyl (**3ha**), trifluoromethyl (**3ia**), cyano (**3ja**) and ester groups (**3ka**) were fully tolerated for this transformation, delivering the desired 4-alkylidene chroman-2-ols smoothly. It should be noted that, for *N*-phenoxyacetamides bearing electron-withdrawing groups (*e.g.* $-CF_3$, –CN, $-CO_2Me$), part of the starting material could be recovered under the standard conditions, which is probably due to the strong electronic properties of such substrates and accounts for the relatively low yields of the corresponding products. Further exploration illustrated that *ortho*- and *meta*-substituted *N*-phenoxyacetamides participated in this [3+3] annulation reaction smoothly, among which the C–H metallation occurred at the less hindered site to give the regio-specific products when *meta*-substituted *N*-phenoxyacetamides were employed (**3la–oa**). The naphthalene substrate was also a viable reactant to afford the corresponding 3,4-dihydro-2*H*-benzo[*g*]chromen-2-ol **3pa**. Remarkably, the dopamine and L-tyrosine derived *N*-phenxoyacetamides underwent the desired coupling with **2a** smoothly under the standard conditions, giving an access to 4-alkylidene chroman-2-ol **3qa** and **3ra** in decent yields, implying the profound potential of the developed protocol for the rapid construction of biologically active molecules *via* late-stage C–H modification.

Subsequently, the generality of this transformation was further extended with regard to a variety of α-allenols. Diverse α-allenol substrates bearing different alkyl substituents at the α position were all compatible with the formation of the desired products (**3ab–ad**). Other α-allenols equipped with various alkyl or aryl substituents also participated in this transformation smoothly, giving access to the corresponding chroman-2-ol derivatives **3ae–aj**. The structure of **3ai** was confirmed using X-ray crystallography (CCDC 2084815†).[9] For aryl- or linear alkyl-tethered α-allenol substrates, the 4-alkylidene chromene

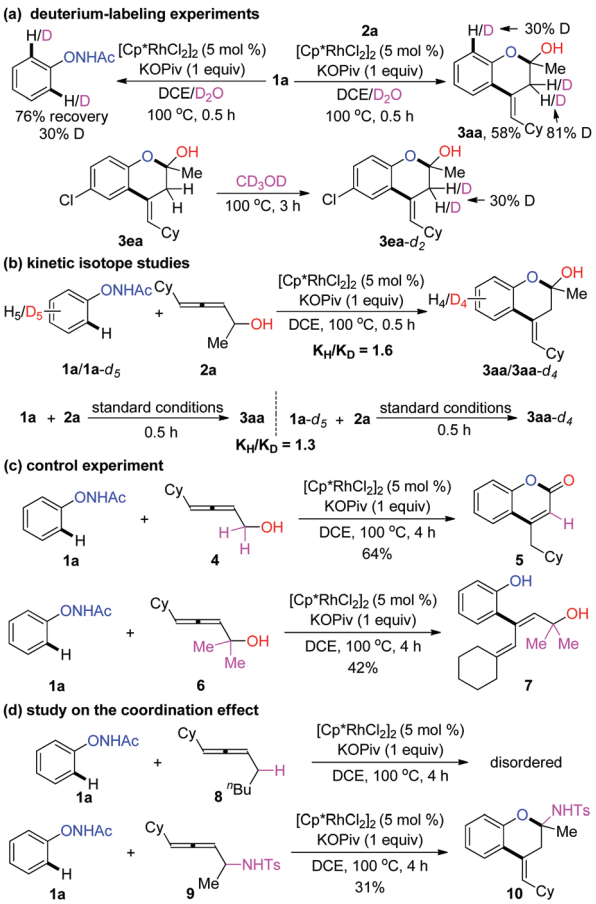


Scheme 2 Substrate scope of *N*-phenoxyacetamides and α-allenols. Reaction conditions: **1a** (0.2 mmol), **2** (0.24 mmol), $[Cp^*RhCl_2]_2$ (5 mol%) and KOPiv (1 equiv.) in DCE (0.2 M) at 100 °C for 4 h without exclusion of air or moisture; isolated yields were reported.

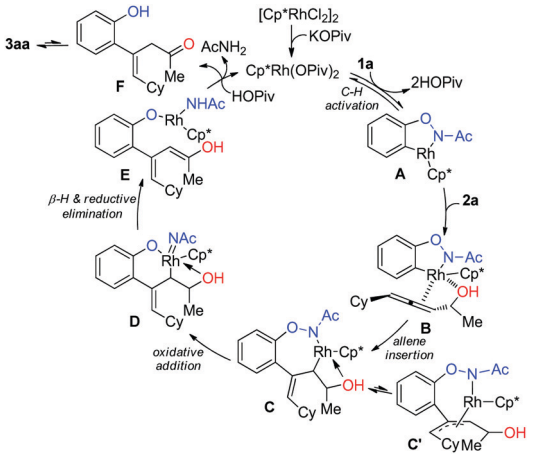
Published on 16 August 2021. Downloaded by Fudan University on 1/17/2022 12:09:40 PM.

skeletons might be formed as the main side-products *via* dehydration of the 4-alkylidene chroman-2-ols, which accounted for the relatively low yields. Of note, only *E*-selective products were obtained regardless of the nature of both the *N*-phenoxyacetamide and α-allenol substrates, revealing the specific stereoselectivity of this reaction. Nevertheless, further examination demonstrated that the α-allenols bearing ester groups, alkenyl or heterocyclic substituents, as well as the tri-substituted α-allenols were not compatible (see the ESI† for details), thus illustrating the substrate limitation of the developed methodology.

To gain more insight into the reaction mechanism and probe the role of the hydroxyl group in tuning the reaction selectivity, a series of experimental mechanistic studies were carried out. Initially, by employing D_2O as the deuterium source, deuterium-labeling experiments were conducted and revealed a reversible C–H metallation. The distinctive deuteration at the allylic position of **3aa** suggested a keto–enol tautomerization process (Scheme 3a). Additionally, the treatment of **3ea** in CD_3OD furnished obvious deuterium incorporation, which suggested an equilibrium between the ring-opening ketone fragment and the chroman-2-ol skeleton bearing a hemiketal moiety. Kinetic isotope studies *via* both the competitive experiment and two parallel, side-by-side reactions revealed the primary KIE values of 1.6 and 1.3, demonstrating



Scheme 3 Experimental mechanistic studies.



Scheme 4 Proposed mechanism.

that the C–H bond cleavage might not be involved in the rate-determining step (Scheme 3b). Moreover, a control experiment using α-allenols **4** and **6** led to the formation of 2*H*-chromen-2-one **5** and diene derivative **7**, respectively, which provided adequate evidence to support that the secondary alcohol moiety was essential in assembling the chroman-2-ol skeleton (Scheme 3c). Further exploration was carried out to verify the coordination effect between –OH and the rhodium metal. As shown in Scheme 3d, the removal of the hydroxyl group resulted in an inseparable mixture of complex products, while the allene substrate bearing a coordinated –NHTs fragment delivered the corresponding 4-alkylidene chroman-2-amide framework **10**. These results illustrated the crucial role of the coordination effect in tuning the specific chemoselectivity to afford chromane products.

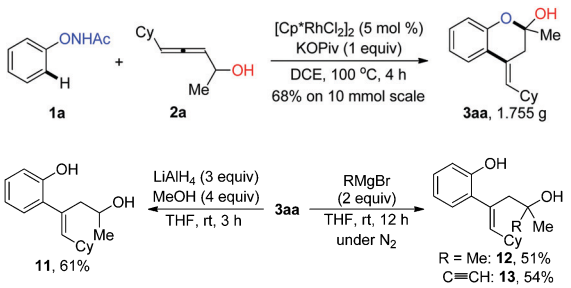
On the basis of the above experimental studies and the literature precedent,[10] a plausible catalytic cycle was proposed for the developed protocol (Scheme 4). With the aid of KOPiv, the active catalyst was generated *via* anion exchange, followed by a reversible C–H bond cleavage to form the five-membered rhodacycle **A**. The following coordination of free hydroxyl groups to rhodium led to the regioselective insertion of an allene double bond into the C–Rh bond, giving the η^1-allylic rhodium intermediate **C**, which was in an equilibrium with the π-allyl rhodium species **C′**. Due to the facile Rh(III)-mediated cleavage of the O–N bond, a Rh(V) species **D** was likely formed *via* oxidative addition.[11] From then on, the β-H elimination then reductive elimination processes occurred smoothly to give the intermediate **E**, which converted into the *ortho*-alkenyl phenol derivative **F** along with the release of $AcNH_2$ and the active Rh(III) catalyst *via* protonolysis. Finally, the cyclization of **F** resulted in the thermodynamically favored chroman-2-ol framework.[12]

In consideration of the synthetic potential, the further scaled-up synthesis of compound **3aa** was explored and revealed that the developed Rh(III)-catalyzed [3+3] annulation could be conducted on a 10 mmol scale without an obvious decrease in yield, implying the practicality of this transformation (eqn (1)). The exposure of **3aa** to $LiAlH_4$ afforded the reductive *ortho*-alkenyl phenol derivative **11**, and the treatment of **3aa** with Grignard reagents led to the formation of

 This journal is © The Royal Society of Chemistry 2021

Published on 16 August 2021. Downloaded by Fudan University on 1/17/2022 12:09:40 PM.

ring-opening products **12** and **13** smoothly, which were equipped with an interesting tertiary alcohol moiety (eqn (2)). These results illustrated the potential utility of the developed approach for the rapid construction of functionalized phenols.



In summary, by synergistically employing –ONHAc and –OH as coordinated DDGs, we have developed the efficient Rh(III)-catalysed C–H [3+3] annulation of *N*-phenoxyacetamides with α-allenols for the specific construction of 4-alkylidene chroman-2-ols. The distinctive coordination effect between free hydroxyl groups and the catalyst metal accounted for the observed regio- and chemoselectivity, thus fulfilling a controllable [3+3] annulation mode using the α-allenol skeleton as an innovative three-carbon annulation component. Further exploration of novel reaction manifolds for allenes in the C–H functionalization field and detailed mechanistic studies are in progress.

We thank the NSFC (21877020, 22007020), the Guangdong Natural Science Funds for Distinguished Young Scholar (2017A030306031) and the Science and Technology Program of Guangzhou (202102020615) for financial support on this study.

Conflicts of interest

There are no conflicts to declare.

Notes and references

1 For selected recent reviews, see: (*a*) D. F. Fernández, J. L. Mascareñas and F. López, *Chem. Soc. Rev.*, 2020, **49**, 7378; (*b*) X.-L. Han, P. Lin and Q. Li, *Chin. Chem. Lett.*, 2019, **30**, 1495; (*c*) S. Yang and M. Shi, *Acc. Chem. Res.*, 2018, **51**, 1667; (*d*) M. Holmes, L. A. Schwartz and M. J. Krische, *Chem. Rev.*, 2018, **118**, 6026; (*e*) Y. Li, X. Yin and M. Dai, *Nat. Prod. Rep.*, 2017, **34**, 1185; (*f*) B. Ye and N. Cramer, *Acc. Chem. Res.*, 2015, **48**, 1308.

2 For selected examples, see: (*a*) R. K. Shukla, A. M. Nair, S. Khan and C. M. R. Volla, *Angew. Chem., Int. Ed.*, 2020, **59**, 17042; (*b*) M. R. Rodríguez, M. Besora, F. Molina, F. Maseras, M. M. Díaz-Requejo and P. J. Pérez, *J. Am. Chem. Soc.*, 2020, **142**, 13062; (*c*) C. Zhu, J. Liu, B. K. Mai, F. Himo and J.-E. Bäckvall, *J. Am. Chem. Soc.*, 2020, **142**, 5751; (*d*) R. Mei, X. Fang, L. He, J. Sun, L. Zou, W. Ma and L. Ackermann, *Chem. Commun.*, 2020, **56**, 1393; (*e*) S.-G. Wang, Y. Liu and N. Cramer, *Angew. Chem., Int. Ed.*, 2019, **58**, 18136; (*f*) C. Ghosh, P. J. Nagtilak and M. Kapur, *Org. Lett.*, 2019, **21**, 3237; (*g*) R. Boobalan, R. Santhoshkumar and C.-H. Cheng, *Adv. Synth. Catal.*, 2019, **361**, 1140; (*h*) X. Vidal, J. L. Mascareñas and M. Gulías, *J. Am. Chem. Soc.*, 2019, **141**, 1862; (*i*) T. H. Meyer, J. C. A. Oliveira, S. C. Sau, N. W. J. Ang and L. Ackermann, *ACS Catal.*, 2018, **8**, 9140; (*j*) C. Lei, L. Peng and K. Ding, *Adv. Synth. Catal.*, 2018, **360**, 2952; (*k*) R. Kuppusamy, R. Santhoshkumar, R. Boobalan, H.-R. Wu and C.-H. Cheng, *ACS Catal.*, 1880, **2018**, 8; (*l*) D.-S. Kong, Y.-F. Wang, Y.-S. Zhao, Q.-H. Li, Y.-X. Chen, P. Tian and G.-Q. Lin, *Org. Lett.*, 2018, **20**, 1154; (*m*) T. Lan, L. Wang and Y. Rao, *Org. Lett.*, 2017, **19**, 972; (*n*) N. Casanova, K. P. Del Rio, R. García-Fandiño, J. L. Mascareñas and M. Gulías, *ACS Catal.*, 2016, **6**, 3349; (*o*) H. Wang, B. Beiring, D.-G. Yu, K. D. Collins and F. Glorius, *Angew. Chem., Int. Ed.*, 2013, **52**, 12430.

3 (*a*) R. Zeng, J. Ye, C. Fu and S. Ma, *Adv. Synth. Catal.*, 2013, **355**, 1963; (*b*) N. Casanova, A. Seoane, J. L. Mascareñas and M. Gulías, *Angew. Chem., Int. Ed.*, 2015, **54**, 2374; (*c*) Z. Zhou, G. Liu and X. Lu, *Org. Lett.*, 2016, **18**, 5668; (*d*) P. Miao, H. Wang, L. Liu, W. Chang and J. Li, *Asian J. Org. Chem.*, 2015, **4**, 1050; (*e*) X. Wu, J. Fan, X. Huang and S. Ma, *Org. Chem. Front.*, 2021, **8**, 4432–4437.

4 For selected reviews, see: (*a*) Y. Zhu, X. Zhao, Q. Wu, Y. Chen and J. Zhao, *Acta Phys.-Chim. Sin.*, 2019, **35**, 989; (*b*) X. Jiang, J. Hao, G. Zhou, C. Hou and F. Hu, *Chin. J. Org. Chem.*, 2019, **7**, 1811.

5 For selected examples, see: (*a*) K. Ozols, S. Onodera, L. Wozniak and N. Cramer, *Angew. Chem., Int. Ed.*, 2021, **60**, 655; (*b*) Y.-L. Mei, W. Zhou, T. Huo, F.-S. Zhou, J. Xue, G.-Y. Zhang, B.-T. Ren, C. Zhong and Q.-H. Deng, *Org. Lett.*, 2019, **21**, 9598; (*c*) Y. Zhu, F. Chen, X. Zhao, D. Yan, W. Yong and J. Zhao, *Org. Lett.*, 2019, **21**, 5884; (*d*) X. Wang, Y. Li, T. Knecht, C. G. Daniliuc, K. N. Houk and F. Glorius, *Angew. Chem., Int. Ed.*, 2018, **57**, 5520; (*e*) Y. Wu, Z. Chen, Y. Yang, W. Zhu and B. Zhou, *J. Am. Chem. Soc.*, 2018, **140**, 42; (*f*) X. Wang, T. Gensch, A. Lerchen, C. G. Daniliuc and F. Glorius, *J. Am. Chem. Soc.*, 2017, **139**, 6506; (*g*) Q. Wu, D. Yan, Y. Chen, T. Wang, F. Xiong, W. Wei, Y. Lu, W.-Y. Sun, J. J. Li and J. Zhao, *Nat. Commun.*, 2017, **8**, 14227; (*h*) B. Li, J. Lan, D. Wu and J. You, *Angew. Chem., Int. Ed.*, 2015, **54**, 14008; (*i*) Z. Zhou, G. Liu, Y. Shen and X. Lu, *Org. Chem. Front.*, 2014, **1**, 1161; (*j*) P. Duan, Y. Yang, R. Ben, Y. Yan, L. Dai, M. Hong, Y.-D. Wu, D. Wang, X. Zhang and J. Zhao, *Chem. Sci.*, 2014, **5**, 1574.

6 For selected examples, see: (*a*) G. Zheng, Z. Zhou, G. Zhu, S. Zhai, H. Xu, X. Duan, W. Yi and X. Li, *Angew. Chem., Int. Ed.*, 2020, **59**, 2890; (*b*) J.-L. Pan, C. Liu, C. Chen, T.-Q. Liu, M. Wang, Z. Sun and S.-Y. Zhang, *Org. Lett.*, 2019, **21**, 2823; (*c*) W. Yi, W. Chen, F.-X. Liu, Y. Zhong, D. Wu, Z. Zhou and H. Gao, *ACS Catal.*, 2018, **8**, 9508; (*d*) W. Chen, F.-X. Liu, W. Gong, Z. Zhou, H. Gao, J. Shi, B. Wu and W. Yi, *Adv. Synth. Catal.*, 2018, **360**, 2470; (*e*) W. Zhou, Y.-L. Mei, B. Li, Z.-Y. Guan and Q.-H. Deng, *Org. Lett.*, 2018, **20**, 5808; (*f*) Y. Wang, Y. Chen, Y. Yang and B. Zhou, *Org. Chem. Front.*, 2018, **5**, 1844.

7 (*a*) M. Wu, S. Lin, G. Zhu, M. Sun, Z. Zhou, H. Gao and W. Yi, *Org. Lett.*, 2020, **22**, 1295; (*b*) M. Bian, L. Ma, M. Wu, L. Wu, H. Gao, W. Yi, C. Zhang and Z. Zhou, *ChemPlusChem*, 2020, **85**, 388.

8 (*a*) A. Nishina, A. Miura, M. Goto, K. Terakado, D. Sato, H. Kimura, Y. Hirai, H. Sato and N. Phay, *Biol. Pharm. Bull.*, 2018, **41**, 770; (*b*) C.-F. Chyu, H.-C. Lin and Y.-H. Kuo, *Chem. Pharm. Bull.*, 2005, **53**, 11; (*c*) F. Fontana, M. Raimondi, M. Marzagalli, M. Audano, G. Beretta, P. Procacci, P. Sartori, N. Mitro and P. Limonta, *Free Radical Biol. Med.*, 2020, **160**, 376.

9 CCDC 2084815 contain the supplementary crystallographic data for this paper (ESI†).

10 For the computational mechanistic studies on C–H functionalization involving α-allenols, see: R. Tian, Y. Li and C. Liang, *J. Org. Chem.*, 2019, **84**, 2642.

11 (*a*) Y. Li, H. Chen, L.-B. Qu, K. N. Houk and Y. Lan, *ACS Catal.*, 2019, **9**, 7154; (*b*) S. Liu, X. Qi, L.-B. Qu, R. Bai and Y. Lan, *Catal. Sci. Technol.*, 1645, **2018**, 8; (*c*) S. Vásquez-Céspedes, X. Wang and F. Glorius, *ACS Catal.*, 2018, **8**, 242; (*d*) J.-Q. Wu, S.-S. Zhang, H. Gao, Z. Qi, C.-J. Zhou, W.-W. Ji, Y. Liu, Y. Chen, Q. Li, X. Li and H. Wang, *J. Am. Chem. Soc.*, 2017, **139**, 3537.

12 For selected examples involving tautomeration between the phenol derivatives and the chroman-2-ol skeleton, see: (*a*) R. Wang, J. Han, C. Li, J. Zhang, Y. Liang, T. Wang and Z. Zhang, *Org. Biomol. Chem.*, 2018, **16**, 2479; (*b*) S. Wang, X. Chen, Q. Ao, H. Wang and H. Zhai, *Chem. Commun.*, 2016, **52**, 9454; (*c*) R. S. Lewis, C. J. Garza, A. T. Dang, T. K. A. Pedro and W. J. Chain, *Org. Lett.*, 2015, **17**, 2278.